A Modular and Cost-Effective Framework for Wireless Foundation Models Through World Knowledge Integration

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Outline

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Executive Summary

- Diagnose the "Wireless AI Wall": imbalance, data scarcity, lack of interpretability.
- Present a modular path to wireless foundation models: DenoMAE encoder, Plug-and-Play reasoning, Meta-Task adaptation.
- Outline future scaling via DenoMAE 2.0, DiSC-AMC, and full-stack integration.

Problem Statement

Background & Related Work

- Vision/NLP foundation models (CLIP, MAE, GPT, BERT) demonstrate transfer, few-shot learning, and interpretability.
- Wireless AI research remains siloed: bespoke CNNs/transformers per dataset, brittle to channel shifts [**?**, **?**].
- Need to bridge success of self-supervised and generative models to physical-layer intelligence without replicating their scale.

Wireless AI Pain Points

- **Condition Imbalance:** Accuracy collapses when moving from high to low SNR; retraining per regime is unsustainable.
- **Data Scarcity:** RF captures require hardware labs and expert annotation; labeled corpora lag petabyte-scale unlabeled streams.
- **Interpretability Gap:** AMC is largely black-box, limiting trust and operator acceptance in spectrum operations.

Foundation Model Cost Barrier

Observation

Training GPT/MAE-scale models demands billions of tokens, GPU-years, and curated data pipelines unavailable in RF labs.

- Wireless signals are noisy, hardware-specific, and governed by channel physics.
- Directly scaling to a "Wireless GPT" repeats the cost profile without reusing world knowledge.

Dataset Scale Mismatch

- Vision: ImageNet-22K (>14M labeled), LAION-5B unlabeled images.
- Language: trillions of tokens + synthetic augmentation.
- Wireless: datasets like RadioML/DeepSig (10–100k labeled) + limited public unlabeled IQ.
- Necessitates architectures that thrive on scarce labels and moderate unlabeled corpora.

Problem Statement

- Wireless AI today = isolated task models with narrow operating envelopes.
- Lack of modular abstractions prevents sharing representations or reasoning.
- Deployment requires cost-effective pathways that exploit existing world knowledge rather than brute-force scale.

Wireless-Specific Challenges

- **Noisy physics:** Semantic meaning buried in subtle constellation statistics.
- **Channel diversity:** Multipath, Doppler, hardware offsets create non-stationary distributions.
- **Operational constraints:** Real-time requirements and edge deployment limit memory + compute.

Research Question 0

Can we reach Wireless Physical-Layer Foundation Models (WPFMs) by **approximating** *world knowledge through modular, reusable components instead of giant monoliths?*

Section Overview & Answer

- We define the "Wireless AI Wall": imbalance, scarcity, black-box behavior.
- Success demands three capabilities: generalization (encoder), reasoning (LLM), adaptation (meta-learning).
- Full-stack monolithic training is infeasible; we need modular, cost-aware approximations.
- **Answer to RQ0:** Pursue foundation capabilities through three coordinated modules—representation, reasoning, adaptation—that reuse world knowledge instead of recreating it.

Proposed Framework

Background & Related Work

- GPT/CLIP pipelines follow a pattern: massive self-supervised pretraining + instruction-tuned reasoning + lightweight adapters.
- Wireless counterparts largely skip pretraining; models are retrained per dataset and lack reasoning layers.
- Prior modular ideas (Mixture-of-Experts, NMformer) still bind components tightly; fail to leverage world knowledge stored in LLMs/Vision FMs.

How GPT-Class Models are Built

1. **Stage A — Representation:** Learn universal token embeddings via masked prediction/next-token tasks.
2. **Stage B — Reasoning:** Align model to follow instructions, cite exemplars, and explain outputs.
3. **Stage C — Adaptation:** Apply LoRA, prefixes, or small finetunes for downstream tasks.

- We adopt this recipe for RF by swapping modalities and cost structures.

Modular Architecture Overview

DenoMAE
(Signal Encoding)

Plug-and-Play
(LLM Reasoning)

Meta-Task
(Rapid Adaptation)

- Modules communicate through interpretable interfaces (images, statistics, latent vectors).
- Together they deliver generalization (encoder), reasoning (LLM), and adaptation (meta-learning).

Research Questions

- **RQ1:** Can vision-style self-supervised learning transfer world knowledge into RF representations?
- **RQ2:** Can LLMs, prompted with signal statistics, provide open-set, interpretable classification?
- **RQ3:** Can meta-learning with learnable regularization enable few-shot adaptation across deployments?

System Flow

1. Collect unlabeled IQ/constellation data (incl. CDRF) $\rightarrow$ pretrain DenoMAE.
2. Summarize encoder outputs into statistics $\rightarrow$ prompt LLM for reasoning.
3. Use Meta-Task to adapt encoder or downstream heads with 5–10 labeled samples.

Data & Compute Strategy

- **Encoding:** 10k unlabeled samples; treat noise as modality for efficiency.
- **Reasoning:** Zero-shot prompts leveraging billions-token LLM priors; no RF retraining.
- **Adaptation:** Episodic few-shot tasks on small datasets (MiniImageNet, FC100) as proxies for AMC deployments.

Evaluation Roadmap

- **Representation:** DenoMAE + CDRF stress-test cross-modal transfer.
- **Reasoning:** Plug-and-Play validated on synthetic AMC with SNR sweeps.
- **Adaptation:** Meta-Task benchmarked on standard FSL datasets before translating to wireless tasks.

Metrics & Targets

- **Representation:** Accuracy at -10 dB SNR, data efficiency (unlabeled+ labeled ratios).
- **Reasoning:** One-shot accuracy across SNR range, open-set rejection rate, explanation quality.
- **Adaptation:** 5-way 1-shot/5-shot accuracy, convergence episodes, sample efficiency.
- Align metrics to highlight cost savings relative to monolithic FM training.

Section Overview & Answer

- **Approach:** Mirror GPT pipeline via three modular components tailored to RF physics.
- **Answer:** A cost-effective wireless foundation model emerges when representation, reasoning, and adaptation are decoupled yet composable.
- Subsequent sections provide empirical validation for RQ1–RQ3.

Signal Encoding: DenoMAE & CDRF

Background & Related Work

- Traditional AMC pipelines: supervised CNNs/ViTs on constellations, little reuse across datasets.
- CDRF benchmark showed YOLO/ResNet (pretrained on COCO/ImageNet) transfer to RF spectrograms.
- Need self-supervised encoder that internalizes noise statistics and scales with unlabeled RF data.

Research Question 1

Can vision-based, self-supervised learning effectively transfer world knowledge to the wireless domain?

CDRF as Empirical Foundation

- 39 drone/RC classes, 500+ GB raw IQ, controlled SNR and interferer augmentations.
- YOLOv11n achieves mAP@0.5 up to 0.986 on clean spectrograms, validating cross-modal transfer.
- Provides realistic noise/interference motifs that motivate noise-aware pretraining.

DenoMAE Overview

- Multimodal masked autoencoder tailored to RF constellations and IQ-derived images.
- Treats **noise** as an explicit modality to disentangle channel effects.
- Encoder is reused; decoders discarded post-pretraining for efficient deployment.

Architecture Pipeline

1. Patchify 224×224 inputs (clean/noisy constellations, clean/noisy signals, noise maps).
2. Mask 75% of patches; process visibles via 12-layer ViT encoder ($d_{model} = 768$).
3. Project shared latent into modality-specific 4-layer decoders to reconstruct masked content.

Pretraining Strategy

- 10k unlabeled samples covering 10 modulation formats (BPSK, QPSK, 8PSK, QAM, GFSK, etc.).
- Masking enforces global reasoning, lowering compute yet boosting denoising skill.
- Optimized with AdamW, batch 64, learning rate 10^{-4}, 100 epochs.

Data Pipeline

- IQ samples reshaped to 32×32, interpolated, replicated to RGB for ViT compatibility.
- Noise modality generated via exponential decay model capturing neighborhood energy.
- Aligns with CDRF's spectrogram evidence, creating an end-to-end cross-modal story.

Results: Robustness

- $+22.1\%$ accuracy at -10 dB SNR relative to non-pretrained ViT baseline.
- Maintains 77.5–84.3% accuracy from -10 to $+10$ dB using only 1k labeled samples.
- Extrapolates down to -20 dB despite training $\geq$ -10 dB (see Figure **??**).

Results: Efficiency

- Needs 10× fewer unlabeled samples than AlexNet AMC and 3× fewer labeled samples at fine-tune.
- Ablations show noise modality delivers 2% extra accuracy, highlighting benefit of treating noise as data.
- Outperforms CNN-AMC, NMformer, DL-GRF on accuracy while using smaller datasets.

Ablation Highlights

- Modalities removed sequentially: still 81.3% with constellations only, 83.5% with all five.
- Clean vs noisy constellation pairing adds +0.6%; adding time-domain signal adds +1.3%.
- Explicit noise branch regularizes encoder during pretraining without affecting inference cost.

Section Summary & RQ1 Answer

- CDRF proves cross-modal feasibility; DenoMAE operationalizes self-supervised transfer.
- Noise-as-modality + heavy masking deliver low-SNR robustness and vast data efficiency while feeding downstream modules.
- **Answer:** RQ1 is affirmed—vision-based self-supervised learning can approximate world knowledge for RF, yielding a reusable encoder for wireless foundation models.

LLM Reasoning: Plug-and-Play

Background & Related Work

- AMC models provide logits without human-readable explanations.
- Open-set AMC is largely unsolved—unknown modulations are forced into nearest class.
- LLM prompting successes (MedPrompt, Toolformer) suggest instructions + exemplars can steer reasoning.

Research Question 2

Can LLMs enable open-set classification of wireless signals through world knowledge transfer?

Plug-and-Play Pipeline

1. Extract descriptive + higher-order statistics from IQ samples.
2. Linearize stats into deterministic text strings.
3. Build prompt = Instruction (I) + exemplars (C) + statistics + answer set (S).
4. Query LLM (DeepSeek, o3-mini) and parse predicted modulation.

Dataset & Experimental Setup

- Synthetic dataset: 10 modulations (4ASK, 4/16PAM, CPFSK, DQPSK, GFSK, GMSK, OOK, OQPSK).
- SNR sweep: noiseless (100 dB) plus [-10,10] dB with random noise injections.
- 20 exemplar samples per condition; remainder used as queries.
- Evaluate multiple LLM sizes (7B, 32B, 200B) and prompt variants ($I + S$ vs $I + C + S$).

Signal Summarization Details

- Compute mean, variance, skewness, kurtosis, moments 0–7, cumulants, k-stats.
- Algorithm **??** ensures stable ordering for reproducible prompts.
- Exemplars pulled from same SNR bucket provide in-context grounding.

Prompt Engineering

- Instruction block restricts answer to known modulations + “Unknown”.
- Context block lists exemplar stats + labels to teach associations.
- Query block provides target stats and answer choices; deterministic decoding enforces concise outputs.

Results

- DeepSeek-R1 32B with context ($I + C + S$) reaches 53.5% accuracy on noisy [-10,10] dB data.
- o3-mini (200B) attains **72.4%** cleaned accuracy without RF-specific training.
- Adding context improves accuracy up to 40% over instruction-only prompts; larger models benefit more.

Interpretability & Open-Set Behavior

- LLM outputs natural-language rationales referencing statistics (e.g., "high kurtosis indicates 16PAM").
- Can output "Unknown" when stats deviate, providing open-set rejection.
- Enables white-box auditing of wireless decisions.

Limitations & Toward DiSC-AMC

- Prompts are verbose; motivates DiSC-AMC (discretized stats, exemplar pruning, constrained prompting).
- Latency/token cost remain high—need distillation to smaller LLMs or on-device caches.
- Future prompts will encode SNR/channel metadata for further robustness.

Section Summary & RQ2 Answer

- Plug-and-Play demonstrates training-free AMC with interpretable reasoning.
- Statistical summaries act as bridge language between RF physics and LLM knowledge.
- **Answer:** Yes—LLMs provide open-set, explainable classification when prompted with RF statistics, fulfilling the reasoning pillar.

Fast Adaption: Meta-Task

Background & Related Work

- Few-shot learning (FSL) methods—MAML, Prototypical Networks, MetaOptNet—still overfit when labeled data per task is tiny.
- Existing regularizers (dropout, weight decay, MetaMix) are static or task-specific.
- Wireless deployments demand fast adaptation to new emitters/SNRs with 5–10 samples.

Research Question 3

Can meta-learning enable rapid adaptation to new tasks with minimal labeled data?

Meta-Task Concept

- Treat regularization itself as a learnable auxiliary task.
- Introduce Task-Decoder that reconstructs inputs from embeddings, forcing rich representations.
- Method-agnostic: drop-in for ProtoNet, MAML, MetaOptNet, P>M>F.

Architecture

- Encoder f_θ shared with base meta-learner.
- Task-Decoder g_ϕ autoencoder reconstructs support+query embeddings.
- Loss: $J = J_{\mathcal{T}} + \lambda J_{\mathcal{M}}$, optimized jointly per episode.

Optimization Dynamics

- Episodes sample support/query; both feed Task-Decoder to learn reconstruction.
- Reconstruction encourages embeddings to retain more than discriminative cues.
- Separate gradients for θ and ϕ maintain stability; no extra labeled data needed.

Datasets & Training Config

- Benchmarks: MiniImageNet, TieredImageNet, FC100, plus transformer-based P>M>F settings.
- Episodes: 5-way 1-shot and 5-way 5-shot, query size 15; 50k episodes per run.
- Task-Decoder variants: shallow (75K params) vs deep (250K) to study capacity trade-offs.
- Optimizers: Adam (1×10^{-4}) for encoder/classifier, 1×10^{-6} for Task-Decoder.

Results Across Backbones

- **ProtoNet:** $+5.9\%$ absolute accuracy on TieredImageNet 5-way 5-shot ($65.4 \rightarrow 71.3$).
- **MetaOptNet:** Improves to 77.3% (MiniImageNet 5-shot) and 78.66% (Tiered) with same training budget.
- **P>M>F:** 92.3% accuracy in 3k episodes vs 90.7% after 4k episodes (faster convergence).

Relevance to Wireless FM

- Rapidly adapt DenoMAE encoders to new modulation/channel conditions with only a handful of labeled traces.
- Task-Decoder can operate on constellation or spectrogram embeddings, making it modality-agnostic.
- Reduces retraining cost for each deployment; aligns with cost-effective foundation vision.

Section Summary & RQ3 Answer

- Meta-Task reformulates regularization as a learnable auxiliary objective, boosting FSL across backbones.
- Demonstrated faster convergence and higher accuracy with minimal labeled data.
- **Answer:** Yes—meta-learning with learnable regularization achieves rapid adaptation, completing the adaptation pillar of the modular framework.

Future Works & Conclusion

Background & Related Work

- Foundation models continue to expand via hybrid generative-discriminative objectives (e.g., MAE 2.0, CLIP 2).
- LLM efficiency research (DiSC, prompt pruning) informs practical deployment.
- Multi-module orchestration (vision-language-action) inspires wireless FM fusion plans.

Future Research Question

How do we scale the three validated modules into an integrated, resource-aware Wireless FM ready for field deployment?

Summary of Contributions

1. **DenoMAE:** Multimodal self-supervised encoder for RF signals (representation pillar).
2. **Plug-and-Play:** LLM-based reasoning for open-set, interpretable AMC (reasoning pillar).
3. **Meta-Task:** Learnable regularization for rapid adaptation (adaptation pillar).

Future Work: DenoMAE 2.0

Hybrid Generative-Discriminative Learning

- Augment reconstruction with position-aware patch classification for spatial awareness.
- Incorporate spectrogram modality from CDRF to capture temporal-frequency cues.
- Goal: learn global + local structure, enabling downstream detection tasks.

Future Work: DiSC-AMC

Token- and Parameter-Efficient LLM Reasoning

- Discretize statistics to compact vocabularies.
- Dynamic exemplar pruning to keep prompts short.
- Constrained prompting to enforce structured outputs on edge devices.

Future Work: Synergistic Integration

- Pipeline: DenoMAE 2.0 features $\rightarrow$ Meta-Task adapters $\rightarrow$ DiSC-AMC reasoning.
- Shared metadata (SNR, channel estimates) passed across modules for context.
- Target: deployable Wireless Physical-Layer FM with interpretable outputs.

Roadmap

- Short term: scale datasets (CDRF expansions), run ablations on DenoMAE 2.0 objectives.
- Mid term: prototype DiSC-AMC on smaller LLMs, integrate with DenoMAE embeddings.
- Long term: system-level demos (spectrum monitoring, adaptive radar) with full modular stack.

Section Summary & RQ Answer

- Future work focuses on enhancing each pillar (representation, reasoning, adaptation) and fusing them.
- **Answer:** Scaling the validated modules through DenoMAE 2.0, DiSC-AMC, and their integration provides a practical path to a deployable wireless foundation model.

Conclusion

This dissertation provides a **blueprint** for the future of Wireless AI. By strategically integrating world knowledge from Vision and Language, we can break the "Wireless AI Wall" and achieve scalable, robust, and intelligent communications.

Thank You Questions?